

MH1101 Tutorial 10 (Week 11)

Solution

Reference: Topic 5.3, 5.4 (Lecture Notes)

1. Determine whether the series is absolutely convergent, conditionally convergent or divergent.

(a) $\sum_{n=0}^{\infty} \frac{(-1)^n}{5n+1}$

(b) $\sum_{n=0}^{\infty} \frac{(-3)^n}{(2n+1)!}$

(c) $\sum_{n=1}^{\infty} \left(\frac{1-n}{2+3n} \right)^n$

(d) $\sum_{n=1}^{\infty} \frac{(-1)^n e^{1/n}}{n^3}$

(e) $\sum_{n=1}^{\infty} \frac{n5^{2n}}{10^{n+1}}$

Solution

(a) Consider the series $\sum_{n=0}^{\infty} a_n$ where $a_n = \left| \frac{(-1)^n}{5n+1} \right| = \frac{1}{5n+1}$. Let $b_n = \frac{1}{n}$. Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1}{5n+1} \cdot \frac{5n}{1} = \lim_{n \rightarrow \infty} \frac{5}{5 + \frac{1}{n}} = \frac{5}{5+0} = 1.$$

So the series $\sum_{n=0}^{\infty} a_n$ diverges by the Limit Comparison Test.

Hence, the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{5n+1}$ is NOT absolutely convergent.

On the other hand, note that $\{a_n\}$ is a decreasing positive sequence that converges to 0:

$$a_{n+1} = \frac{1}{5(n+1)+1} < \frac{1}{5n+1} = a_n, \quad \text{for all } n \geq 0,$$

$$\lim_{n \rightarrow \infty} a_n = 0.$$

By the Alternating Series Test, the series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{5n+1} = \sum_{n=0}^{\infty} (-1)^n a_n$$

converges.

Hence, the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{5n+1}$ converges conditionally.

(b) Let $a_n = \frac{(-3)^n}{(2n+1)!}$

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(-3)^{n+1}}{(2(n+1)+1)!} \cdot \frac{(2n+1)!}{(-3)^n} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{-3}{(2n+3)(2n+2)} \right| \\ &= 3 \lim_{n \rightarrow \infty} \frac{1}{(2n+3)(2n+2)} \\ &= 0 < 1. \end{aligned}$$

By the Ratio Test, the series $\sum_{n=0}^{\infty} a_n$ converges absolutely.

(c) Let $a_n = \left(\frac{1-n}{2+3n}\right)^n$. Then $\sqrt[n]{|a_n|} = \left|\frac{1-n}{2+3n}\right| = \frac{n-1}{2+3n}$ since $n \geq 1$.

$$\begin{aligned} \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} &= \lim_{n \rightarrow \infty} \frac{n-1}{2+3n} \\ &= \lim_{n \rightarrow \infty} \frac{1 - \frac{1}{n}}{\frac{2}{n} + 3} = \frac{1}{3} < 1. \end{aligned}$$

By the Root Test, the series $\sum_{n=1}^{\infty} \left(\frac{1-n}{2+3n}\right)^n$ is absolutely convergent.

(d) Let $a_n = \frac{(-1)^n e^{1/n}}{n^3}$. Note that

$$0 \leq |a_n| = \frac{e^{1/n}}{n^3} \leq \frac{e}{n^3}.$$

Since $\sum_{n=1}^{\infty} \frac{1}{n^3}$ converges (since it is a p -series with $p > 1$), the series $\sum_{n=1}^{\infty} \frac{e}{n^3}$ also converges. By the Comparison Test, the series $\sum_{n=1}^{\infty} |a_n|$ converges, i.e. $\sum_{n=1}^{\infty} a_n$ converges absolutely.

(e) Let $a_n = \frac{n5^{2n}}{10^{n+1}}$.

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{(n+1)5^{2(n+1)}}{10^{n+1+1}} \cdot \frac{10^{n+1}}{n5^{2n}} \\ &= \lim_{n \rightarrow \infty} \frac{5^2}{10} \left(\frac{n+1}{n} \right) \\ &= \frac{5}{2} \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right) \\ &= \frac{5}{2} > 1.\end{aligned}$$

By the Ratio Test, the series diverges.

Alternatively, we can apply the n -Term Test for Divergence:

$$a_n = \frac{n}{10} \left(\frac{5^2}{10} \right)^n \geq \frac{1}{10} \left(\frac{5}{2} \right)^n \quad \text{for all } n \geq 1.$$

Since $\lim_{n \rightarrow \infty} \left(\frac{5}{2} \right)^n = \infty$, we deduce that

$$\lim_{n \rightarrow \infty} a_n = \infty.$$

So by the n -Term Test for Divergence, the series $\sum_{n=1}^{\infty} a_n$ diverges.

□

2. For which positive integers k is the following series convergent?

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{(kn)!}.$$

Solution We use the Ratio Test:

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{((n+1)!)^2}{(k(n+1))!} \cdot \frac{(kn)!}{(n!)^2} \\ &= \lim_{n \rightarrow \infty} \frac{(n+1)^2}{(kn+k)(kn+k-1) \cdots (kn+1)}\end{aligned} \tag{1}$$

- If $k = 1$, then (1) becomes

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{(n+1)^2}{n+1} \\ &= \lim_{n \rightarrow \infty} (n+1) = \infty.\end{aligned}$$

Thus, the series diverges.

- If $k = 2$, it follows from (1) that

$$\begin{aligned}\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow \infty} \frac{(n+1)^2}{(2n+2)(2n+1)} \\ &= \lim_{n \rightarrow \infty} \frac{(n+1)(n+1) \cdot \frac{1}{n^2}}{(2n+2)(2n+1) \cdot \frac{1}{n^2}} \\ &= \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right) \left(1 + \frac{1}{n}\right)}{\left(2 + \frac{2}{n}\right) \left(2 + \frac{1}{n}\right)} \\ &= \frac{1}{4} < 1.\end{aligned}$$

So by the Ratio Test, the series converges (absolutely).

- If $k \geq 3$, note that

$$\begin{aligned}0 \leq \frac{(n+1)^2}{(kn+k)(kn+k-1) \cdots (kn+1)} &\leq \frac{(n+1)^2}{(kn+3)(kn+2)(kn+1)} \\ &\leq \frac{(n+1)^2}{(3n+3)(3n+2)(3n+1)}\end{aligned}$$

and

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{(n+1)^2}{(3n+3)(3n+2)(3n+1)} &= \lim_{n \rightarrow \infty} \frac{(n+1)^2 \cdot \frac{1}{n^3}}{(3n+3)(3n+2)(3n+1) \cdot \frac{1}{n^3}} \\ &= \lim_{n \rightarrow \infty} \frac{\frac{1}{n} \left(1 + \frac{1}{n}\right) \left(1 + \frac{1}{n}\right)}{\left(3 + \frac{3}{n}\right) \left(3 + \frac{2}{n}\right) \left(3 + \frac{1}{n}\right)} \\ &= \frac{0 \cdot 1 \cdot 1}{3 \cdot 3 \cdot 3} = 0.\end{aligned}$$

Thus, by Squeeze Theorem,

$$\lim_{n \rightarrow \infty} \frac{(n+1)^2}{(kn+k)(kn+k-1) \cdots (kn+1)} = 0.$$

It follows from (1) that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 0.$$

So by the Ratio Test, the series converges (absolutely). □

3. Prove that $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$ for all x .

Hint: Use the Ratio Test.

Solution Let $a_n = \frac{x^n}{n!}$. Then

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} = \lim_{n \rightarrow \infty} \frac{x}{n+1} = 0 < 1.$$

By the Ratio Test, the series $\sum_{n=1}^{\infty} \frac{x^n}{n!}$ is convergent. The terms of a convergent series must converge to 0 (the n th Term Test for Divergence). Thus,

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0. \quad \square$$

4. Prove the Root Test:

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence and assume that the following limit exists:

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

- (i) If $L < 1$, then $\sum a_n$ converges absolutely.
- (ii) If $L > 1$ or $L = \infty$, then $\sum a_n$ diverges.
- (iii) If $L = 1$, the Root Test is inconclusive. The series may converge or diverge.

Solution

- (i) Take any r such that $L < r < 1$. Since $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L$, for any $\varepsilon > 0$, there exists N such that

$$n > N \implies |\sqrt[n]{|a_n|} - L| < \varepsilon.$$

Choose $\varepsilon = r - L$. Then there exists N such that

$$|\sqrt[n]{|a_n|} - L| < r - L \quad \text{for all } n > N$$

$$-(r - L) < \sqrt[n]{|a_n|} - L < (r - L) \quad \text{for all } n > N.$$

We are interested in the upper bound:

$$\sqrt[n]{|a_n|} < r \quad \text{for all } n > N.$$

This implies that

$$|a_n| < r^n, \quad \text{for all } n > N.$$

Since the geometric series $\sum_{n=N+1}^{\infty} r^n$ converges (since $|r| < 1$), by the Comparison Test, we deduce that the series $\sum_{n=N+1}^{\infty} |a_n|$ converges.

It follows that the series $\sum_{n=1}^{\infty} |a_n|$ converges, i.e. the series $\sum_{n=1}^{\infty} a_n$ converges absolutely.

- (ii) Suppose $L \in \mathbb{R}$ and $L > 1$. Choose any r such that $L > r > 1$. Since $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L$, for any $\varepsilon > 0$, there exists N such that

$$n > N \implies |\sqrt[n]{|a_n|} - L| < \varepsilon$$

Choose $\varepsilon = L - r$. Then there exists an integer N such that

$$\begin{aligned} |\sqrt[n]{|a_n|} - L| &< L - r \quad \text{for all } n > N \\ -(L - r) &< \sqrt[n]{|a_n|} - L < L - r \quad \text{for all } n > N \end{aligned} \quad (2)$$

We are interested in the lower bound:

$$\sqrt[n]{|a_n|} > r \quad \text{for all } n > N.$$

This implies that

$$|a_n| > r^n > 1 \quad \text{for all } n > N.$$

It follows that $\lim_{n \rightarrow \infty} a_n \neq 0$. So by the n -th Term Test for Divergence, the series $\sum_{n=1}^{\infty} a_n$ diverges.

The case for $L = \infty$ is similar. If we choose any $r > 1$, then since $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \infty$, eventually $\sqrt[n]{|a_n|}$ will be greater than r , i.e. (2) will hold. And the rest of the argument is the same.

- (iii) Consider the series $\sum_{n=1}^{\infty} \frac{1}{n}$ which diverges and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ which converges.

We check $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$ for each series.

- For $\sum_{n=1}^{\infty} \frac{1}{n}$,

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \frac{1}{n^{1/n}} = 1. \quad (\text{see Tutorial 7 Question 4(x)})$$

- For $\sum_{n=1}^{\infty} \frac{1}{n^2}$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \frac{1}{(n^{1/n})^2} = 1.$$

Both series have $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1$, but one of them is convergent whereas the other is divergent. $\square$

5. Let $\sum_{n=1}^{\infty} a_n$ be a series with positive terms, and let $r_n = \frac{a_{n+1}}{a_n}$. Suppose $\lim_{n \rightarrow \infty} r_n = L < 1$.

Let $R_n = a_{n+1} + a_{n+2} + \cdots$. If $\{r_n\}$ is a decreasing sequence and $r_{n+1} < 1$, show that

$$R_n \leq \frac{a_{n+1}}{1 - r_{n+1}}.$$

Solution

$$R_n = a_{n+1} + a_{n+2} + a_{n+3} + a_{n+4} \cdots \quad (3)$$

$$= a_{n+1} \cdot \left(1 + \frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+3}}{a_{n+1}} + \frac{a_{n+4}}{a_{n+1}} + \cdots \right) \quad (4)$$

$$= a_{n+1} \cdot \left(1 + \frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+3}}{a_{n+2}} \frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+4}}{a_{n+3}} \frac{a_{n+3}}{a_{n+2}} \frac{a_{n+2}}{a_{n+1}} + \cdots \right) \quad (5)$$

$$= a_{n+1} \cdot (1 + r_{n+1} + r_{n+1}r_{n+2} + r_{n+1}r_{n+2}r_{n+3} \cdots) \quad (6)$$

Now, since $\{r_n\}$ is decreasing, we have $r_{n+i} \leq r_{n+1}$ for all $i = 2, 3, 4, \dots$. It follows from (6) that

$$R_n \leq a_{n+1} \cdot (1 + r_{n+1} + r_{n+1}^2 + r_{n+1}^3 + \cdots)$$

Notice that the series $1 + r_{n+1} + r_{n+1}^2 + r_{n+1}^3 + \cdots$ is a geometric series with first term 1 and common ratio $r_{n+1} < 1$, so it converges to

$$\frac{1}{1 - r_{n+1}}.$$

Hence,

$$R_n \leq \frac{a_{n+1}}{1 - r_{n+1}}.$$

$\square$

Answer Keys.

1. (a) converges conditionally (b) converges absolutely (c) converges absolutely (d) converges absolutely (e) diverges **2.** diverges for $k = 1$, converges (absolutely) for $k \geq 2$